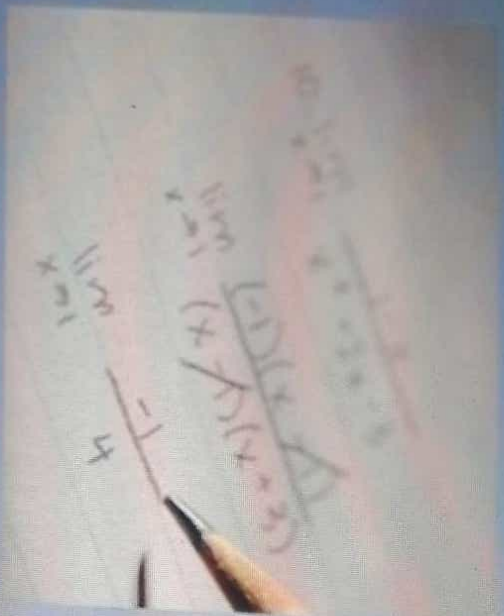


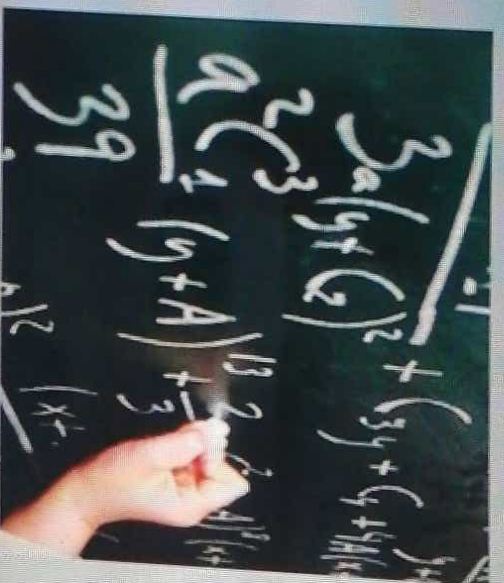
USING BML IN CONFIGURATION

BML - OVERVIEW

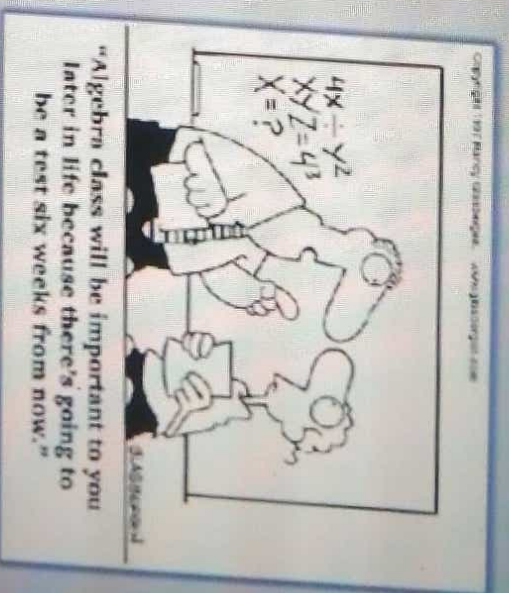
BigMachines Markup Language (BML) is a scripting tool that is used to define advanced login in configuration and commerce. Anywhere you see a "Define Function" button, you will open a BML function editor.



Handwritten algebraic equations on a piece of paper, including a fraction $\frac{(x-1)(x+1)}{(x+1)(x+3)}$ and a subtraction $\frac{-1}{4}$.



Handwritten algebraic equations on a chalkboard, including $\frac{3a(y+6)^2 + (3y+4+4y)}{a^2c^3(y+4) + \frac{2}{3}y^2 + 4y^3}$.



Algebraic equations use combinations of **variables** and **functions** to solve for one or more variables. Similar in theory, BML also uses variables and functions to capture complex business logic.

BML - BASICS

Variables:

Variables are placeholders for data. Some variables are given(inputs) and others are dependent variables(outputs), meaning their value is dependent on the value of the given variables.

Functions:

A function is usually defined as a rule that relates one variable to another. Functions using input variables to produce the value of an output variable.

Sample BML Script

```
1 strVar_1="123";  
2 strVar_2="";  
3  
4 floatResults=atoi(strVar_1);  
5     print floatResults;  
6  
7  
8 integerResults=atoi(strVar_1);  
9     print integerResults;  
10  
11 return floatResults;
```

Input Variables

Function

Output Variables

In this example, we are using functions with the input variables to get the value of the output variable.

BML - BASICS

Basic Data Types

A data type determines the values that a variable can contain and how it is executed.

Integer



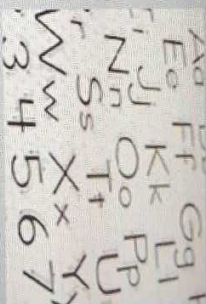
Float



Boolean



String



String Variables must be enclosed in **double-quotations** (" "). Numerical values and **boolean** values do not need to be enclosed in quotations.

BML - BASICS

Common Elements of a BML Script

Element	Description
;	Represents an end statement. This is required on each line of code.
//	Indicates a comment line.
return	This is the beginning of the return statement which is required at the end of your code. BML will not process instructions without it.
print	This function displays all data types in the console to make debugging easier. You can give it direct inputs or have it print variables and results from other functions.
=	Use to assign a value to a variable.

```
1 //Basic syntax of BML
2
3 age = "32";
4 salary = "52,021.62";
5 middleInitial = "J.";
6
7 results = "My name is John " + middleInitial + " Smith and I earn " +
8         salary + " per year. I am " + age + " years old.";
9
10 return results;
```

23

© 2010 Dedeitte Touche Tomatsu

BML – FUNCTION EDITOR

The Function Editor is where you will go to write your advanced rules. It's basically the same throughout BigMachine, with only slight variations depending on where you are (Commerce, Configuration, Util Library etc.). The function editor can be accessed by clicking "Advanced Function" in any rule. It can also be accessed by going to the BML Library from the Admin Home Page.

Developer Tools

Administration Logs
BML Library
Data Tools
Error Logs
Global Search on BML

Click to access **Function Editor** from either the Admin Home Page OR within a rule.

Properties: New Rule

Name:

Variable Name:

Description:

Mandatory:

Condition: (This rule always fires, and returns true)

Condition Type:

Always True

Define Logic

Active: (Defines Recommended Items)

Recommended Items rule enables you to associate entire sets of parts and nodes with products based on buyer-configured values.

Comments:

System recommended item

Simple

Advanced

Define Function

Legacy Format (Item1~quantity1~item2~quantity2)

With Advanced Price and Comments (Item1~quantity1~comment1~price1 | ~Item2~quantity2~comment2~price2)

Recommended Items >> Product

Status:

Active

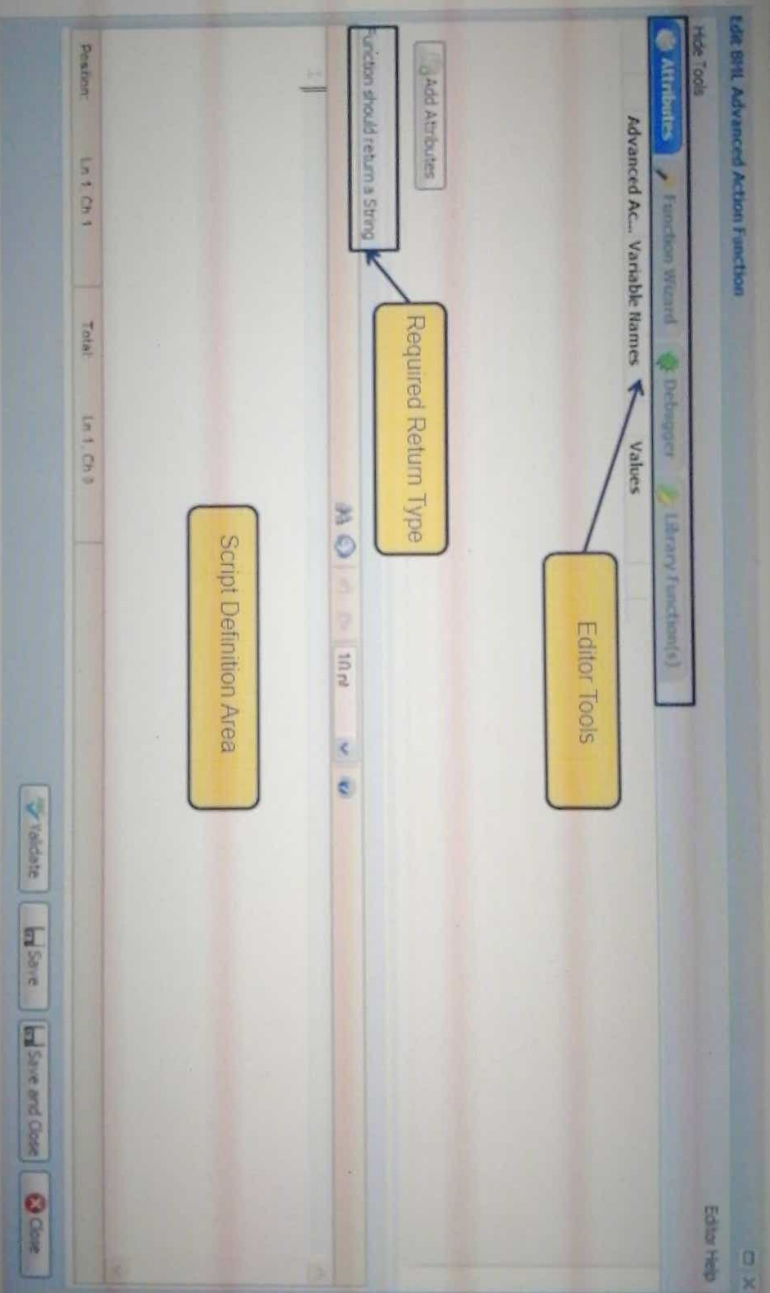
Inactive

Deactive

Edit Start/End Dates

BML – FUNCTION EDITOR

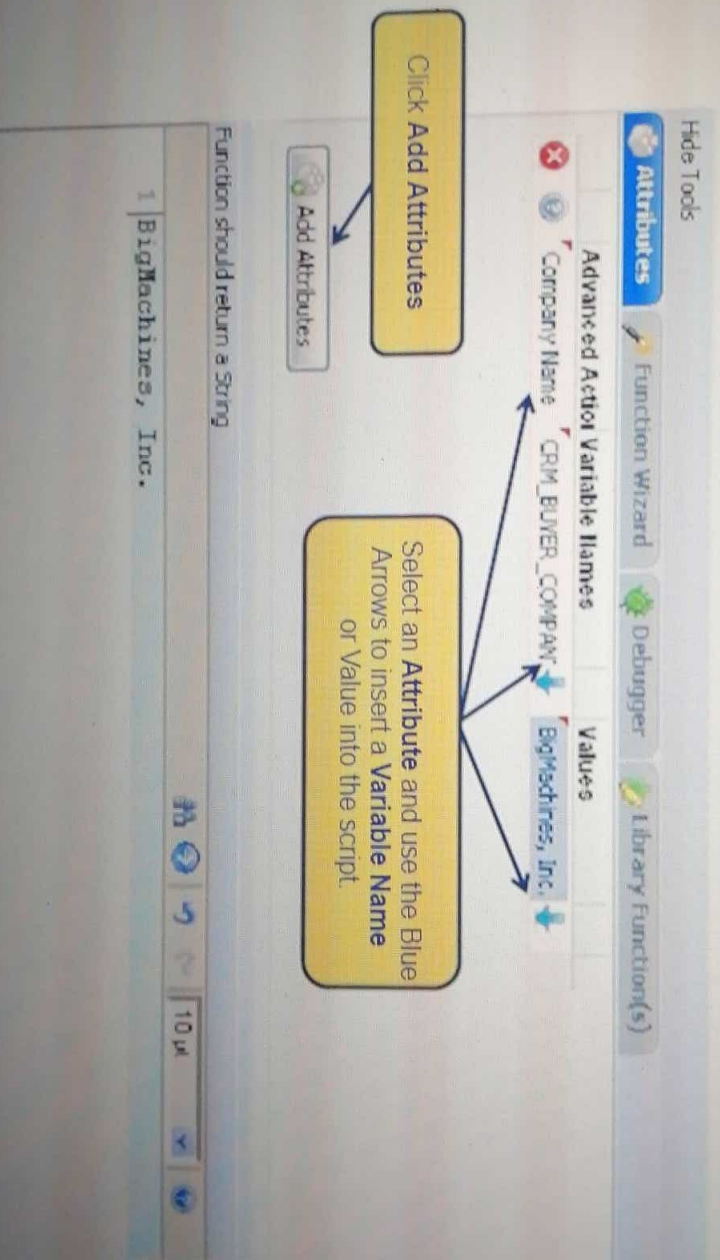
Sample Function Editor



BML – FUNCTION EDITOR

Attributes and Actions

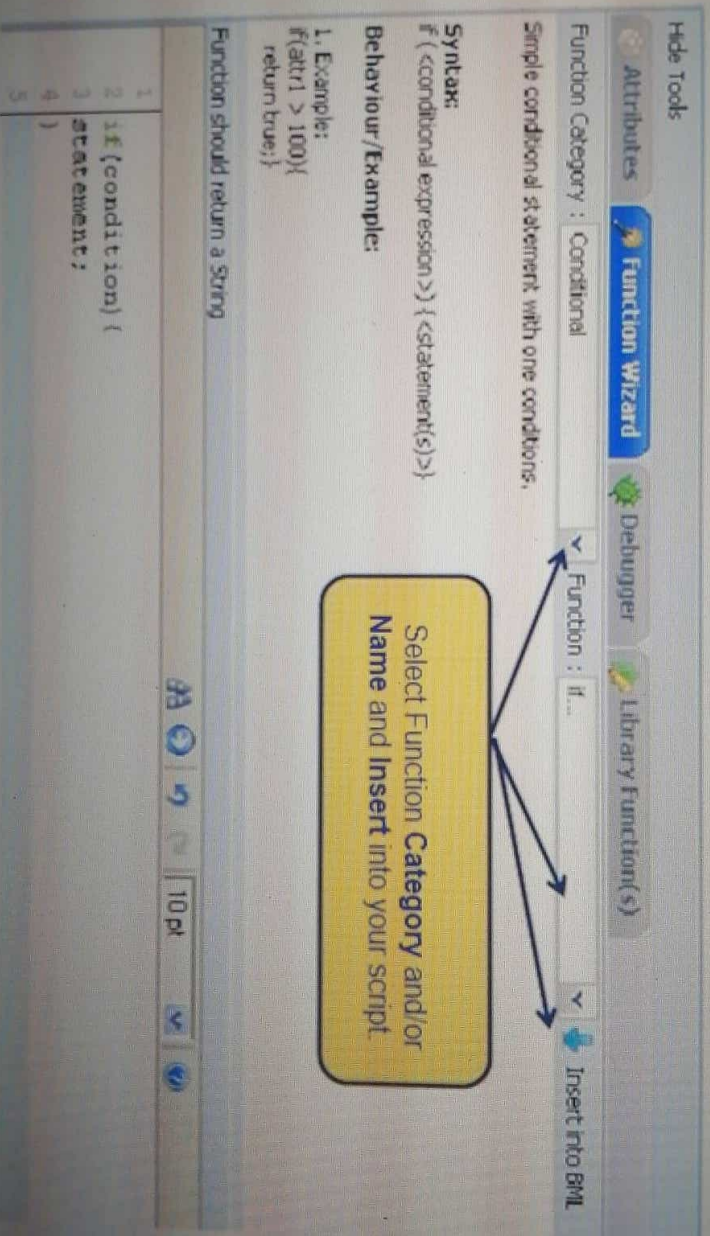
In the Function Editor, you can select attributes to include in your script. These attributes are specific to the product family you are working in.



BML – FUNCTION EDITOR

Function Wizard

The Function Wizard contains a list of pre-defined functions that are available for use in advanced scripting. When you have selected a function, a description of what it does, the proper syntax and an example will be displayed to help you determine if it's what you need.



BML – FUNCTION EDITOR

Debugger

The Debugger allows you to test your script for errors and to ensure it is returning the values you need.

The screenshot shows the BML Function Editor interface. The top bar includes tabs for 'Code Tools', 'Attributes', 'Function Wizard', and 'Debugger'. The 'Debugger' tab is active. Below the tabs are sections for 'Test Script', 'Params', and 'Cannon'. The 'Test Script' section has a checkbox labeled 'Use Test Script'. The main area is a code editor with the following script:

```
1 salary = "52000.62";  
2 middleinitial = "J.";  
3  
4 results = "My name is John " + middleinitial + " Smith and I earn " +  
5 salary + " per year. I am " + age + " years old.";  
6  
7  
8  
9  
10  
11  
12 return results;
```

Below the code editor is a console window showing the output: "My name is John J Smith and I earn 52000.62 per year. I am 32 years old." The text is color-coded: "My name is John J" is purple, "Smith and I earn 52000.62 per year. I am 32 years old." is yellow, and the entire line is highlighted in yellow.

A callout box with a yellow background and black border contains the following text:

Notice the Purple text and the Yellow highlighted text.
The purple text represents the "print" statement and the
Yellow highlighted text represents the return statement

At the bottom right of the code editor, there are three buttons: 'Expand Console', 'Clear', and 'Run'. A yellow callout box with a black border and an arrow pointing to the 'Run' button contains the text:

Click Run to test your script in the Debugger.

BML - PRICING RULE

What do we want to happen?

When the user selects the size and crust type of their pizza, the price for that pizza is calculated and displayed

How can we make it happen?

Write a pricing rule that will evaluate the selections of both those attributes, and return the correct price associated with it.